EEEE4138 Aerial Robotics
Tutorial 5 Flight Control II

Flight Control II Tutorial

Cascaded control and state feedback



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Conventions

Use g = 9.81 m/s² and radians in calculations.

Downwards z is positive. Positive pitch is nose-up.

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Question 1 Two connected loops

A level quadrotor must move towards a target ahead.

(a) State the roles and outputs of the outer and inner loops.

(b) Explain why horizontal movement requires tilt.

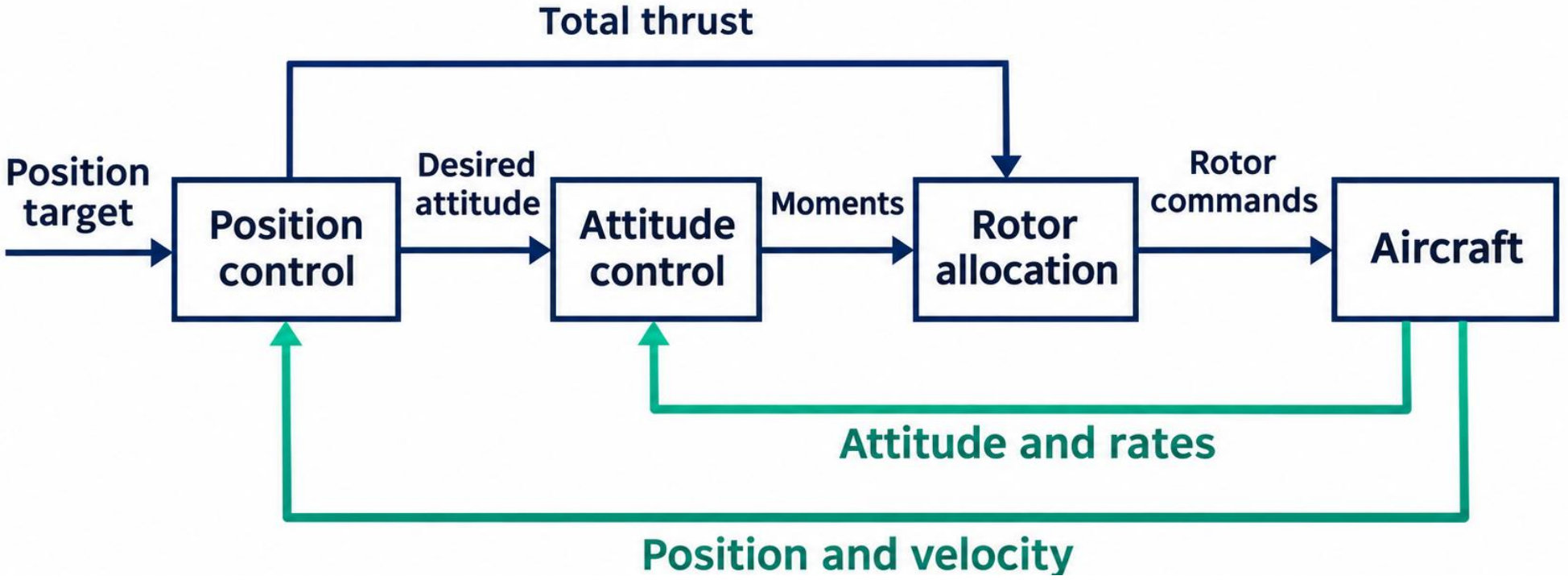
(c) Explain why the inner loop should respond faster.

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Solution 1 Position and attitude control

Outer loop uses position/velocity to request acceleration and tilt.

Inner loop uses attitude/rates to request turning moments.



Tilt redirects thrust. A faster attitude loop supports position tracking.

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Question 2 From position error to pitch

Near hover at zero heading, neglect drag and coupling.

$e_x = +0.0400$ m, $\dot{e}_x = -0.0250$ m/s, $\ddot{x}^d = 0$.

Use $k_{px} = 9.00\ s^{-2}$ and $k_{dx} = 4.80\ s^{-1}$.

(a) Calculate u_x and θ^d using $\theta^d = -u_x/g$.

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Solution 2a Horizontal acceleration and tilt

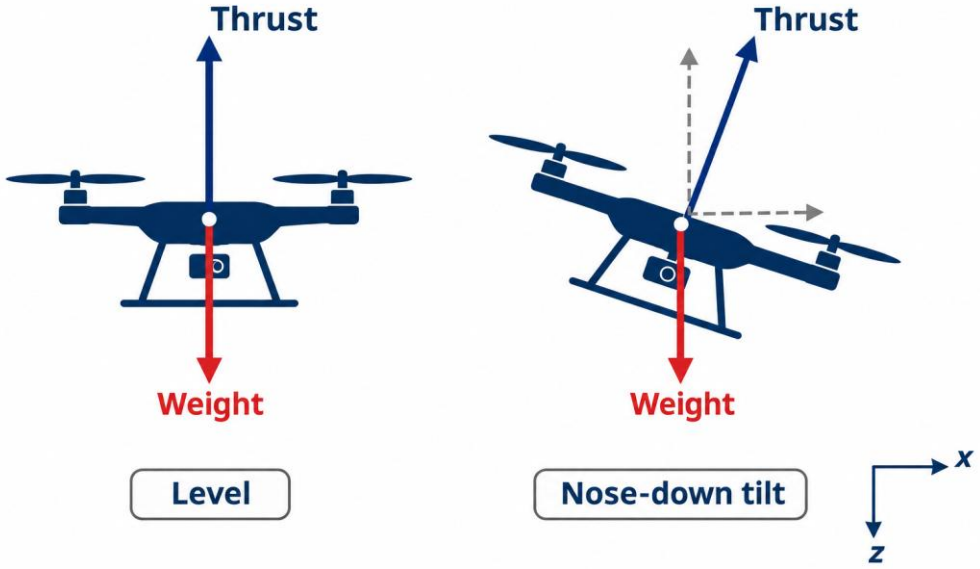
$$u_x = \ddot{x}^d + k_{px}e_x + k_{dx}\dot{e}_x$$

$$u_x = 0 + 0.360 - 0.120$$

$$u_x = +0.240 \text{ m/s}^2$$

$$\theta^d = -0.240/9.81$$

$$\theta^d = -0.0245 \text{ rad} \approx -1.40°$$



Negative pitch directs thrust towards +x.

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Question 2b Pitch moment and gain units

At this instant $\theta = 0$ and $q = +0.0500$ rad/s.

Set $\ddot{\theta}_d = \dot{\theta}_d = 0$ for this instantaneous calculation.

$J_Y = 0.00800$ kg m², $k_p\theta = 144$ s⁻², $k_d\theta = 19.2$ s⁻¹.

Calculate α_cmd, M_Y,cmd and equivalent torque gains.

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Solution 2b Angular acceleration to torque

$\alpha_cmd = 144(-0.240/9.81) - 19.2(0.0500)$

$\alpha_cmd = -4.48\ rad/s^2.$

$M_Y,cmd = J_Y\alpha_cmd = -0.0359\ N\ m.$

$K_p\theta = 1.152\ N\ m/rad,\ K_d\theta = 0.1536\ N\ m\ s/rad.$

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Question 3 A vertical state-space model

Use level motion with m = 2.00 kg and no actuator lag.

States $\xi = [\delta z, \delta w]^T$. Input is thrust change δT in N.

$\delta \dot{z} = \delta w$ and $\delta \dot{w} = -\delta T/m$. Measure position deviation.

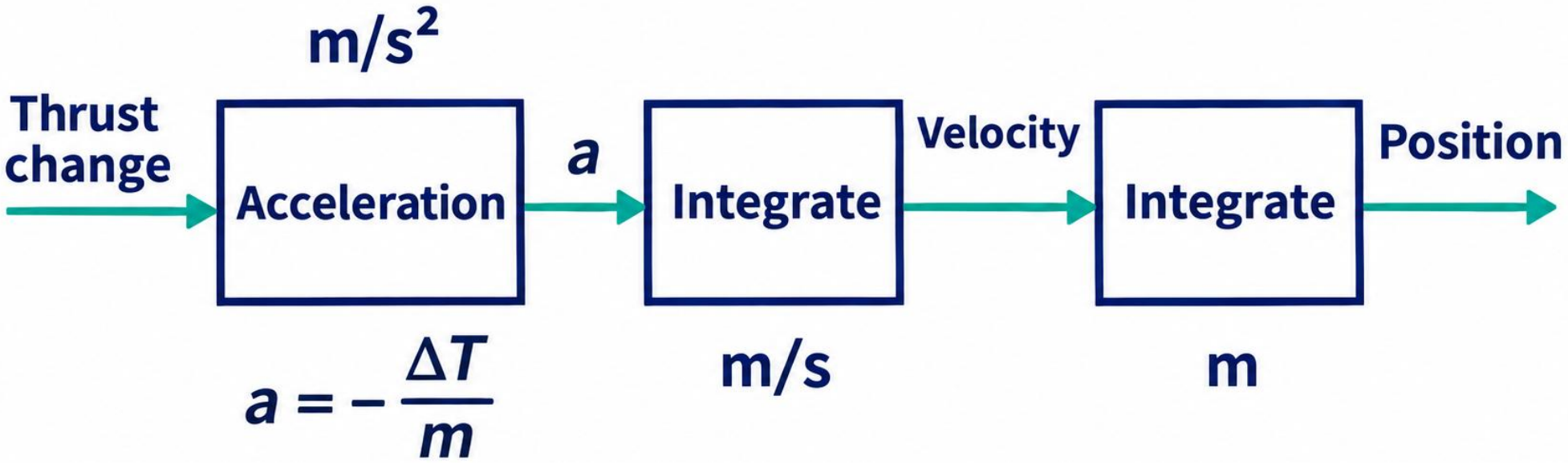
(a) Write A, B_T, C and D. Explain the entries of B_T.

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Solution 3a State and input coefficients

A = [0 1; 0 0] and B_T = [0; −0.500].

C = [1 0] and D = 0.



Thrust changes velocity through −1/m. Position responds through velocity.

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Question 3b Control and observation

Form the controllability matrix [B_T AB_T].

Form the observability matrix [C; CA].

Find both ranks and interpret them.

Does a zero first entry in B_T make position uncontrollable?

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Solution 3b Both states can be reached and inferred

AB_T = [−0.500; 0].

[B_T AB_T] = [0 −0.500; −0.500 0], rank 2.

[C; CA] = [1 0; 0 1], rank 2.

The model is controllable and observable.

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Question 4 Feedback and pole placement

Now choose acceleration input $v_c = -\delta T/m$.

$A = [0\ \ 1; 0\ \ 0]$, $B = [0; 1]$, $v_c = -K\xi$.

(a) Choose K to place poles at -2.00 and $-5.00\ s^{-1}$.

Write $A - BK$ and the closed-loop characteristic polynomial.

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Solution 4a Coefficient matching

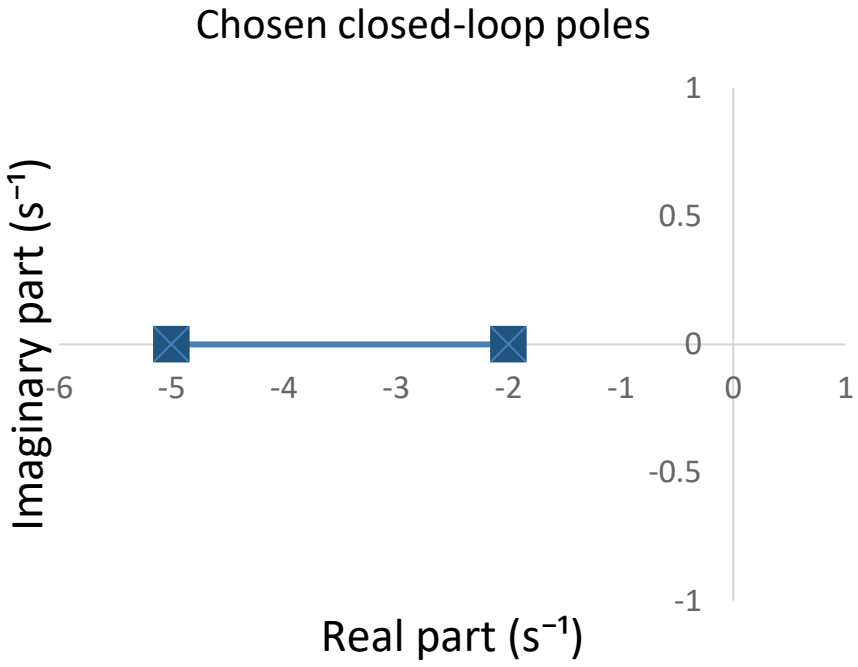
$(s + 2)(s + 5) = s^2 + 7s + 10$

Model: $s^2 + k_2 s + k_1$

$K = [10.0 \;\; 7.00]$

Gain units: s^{-2} and s^{-1}

$A - BK = [0 \;\; 1; -10 \;\; -7]$



Both poles are real and negative.

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Question 4b Reference tracking and physical thrust

Use $v_c = k_1(r - \delta z) - k_2 \delta w$, with the gains just found.

At this instant $r = -0.300$ m, $\delta z = -0.100$ m.

$\delta w = +0.0500$ m/s and $m = 2.00$ kg.

Find v_c, δT and total thrust. Explain the acceleration direction.

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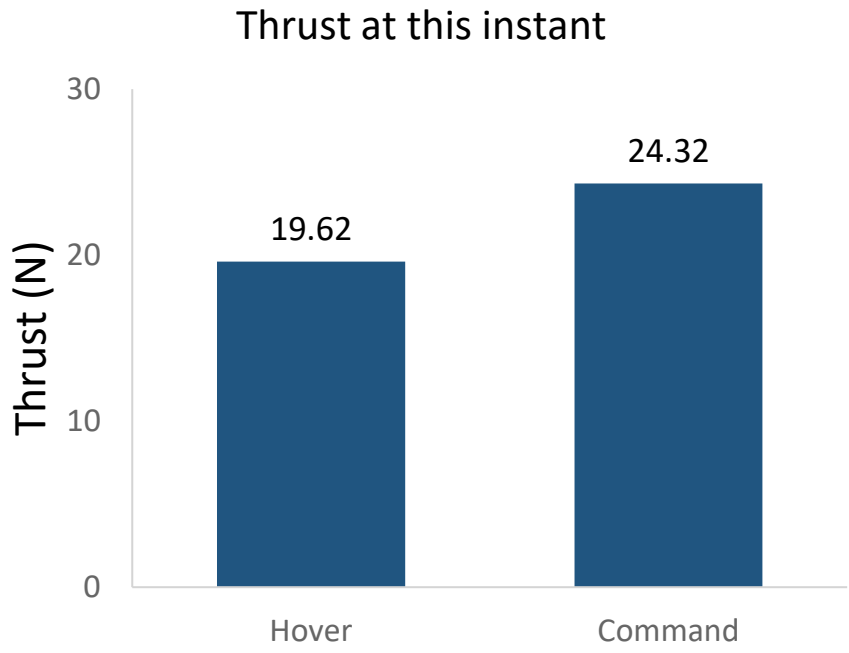
Solution 4b Equilibrium thrust must return

$$v_c = 10(-0.300 + 0.100) - 7(0.0500)$$

$$v_c = -2.35 \text{ m/s}^2, \text{ directed upwards.}$$

$$\delta T = -mv_c = +4.70 \text{ N.}$$

$$T_{cmd} = mg + \delta T = 19.62 + 4.70 = 24.32 \text{ N}$$



Extra thrust gives upward acceleration.
Downwards z remains positive.

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Question 5 PD and state feedback

Compare $u_z = k_p(z^d - z) - k_d\dot{z}$ with the previous law.

(a) State when these two laws are equivalent.

(b) Must state feedback always give better tracking? Explain.

(c) Does state feedback always require outer and inner loops?

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Solution 5 The control law and its arrangement

For constant targets, $r - \delta z = z^d - z$ and $\delta w = \dot{z}$.

The laws coincide when $k_1 = k_p$ and $k_2 = k_d$.

Equivalent laws and implementations can give identical responses.

State feedback may use cascaded channels or one combined controller.

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Final discussion

Why does a forward request create a negative pitch command?

Why does thrust input have a different B from acceleration input?

Why must mg be included in the physical thrust command?

What extra state appears when an actuator has lag?

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